Kajal Avanthika KM

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.74
The Suguna PIP School Coimbatore (TN)	Class XII (CBSE)	2022	94.60%
Kamala Niketan Montessori School Trichy (TN)	Class X (CBSE)	2020	95.40%

WORK EXPERIENCE

Data Analyst Intern | American Express

Gurgaon, India

- **Engineered predictive risk & marketing features** from credit bureau SORs, response histories, and risk metrics.
- **Pruned feature redundancy** using IV and SHAP values to isolate top drivers of cardholder engagement.
- **Built SQL extraction pipelines on GCP** across training and scoring environments to eliminate train-serve skew.
- **Trained and tuned an XGBoost classifier on 1.5M+ cardmember records**.
- **Delivered a 4% relative Gini uplift** over baseline models for personalized offer targeting.
- **Validated model stability and explainability** across multiple portfolios and Out-of-Time (OOT) datasets.

PROJECTS

Image Classification CNN in PyTorch

Built and trained a Convolutional Neural Network (CNN) using PyTorch to classify multi-class image datasets (CIFAR-10) with customized neural network architecture, activation functions, and optimizer pipelines.

Multivariate Time Series Weather Forecasting

Developed and evaluated LSTM, GRU, and 1D-CNN architectures to predict multivariate environmental metrics (temperature and pressure), implementing feature engineering and target standard scaling to optimize multi-output loss functions.

Decoding Airfare Volatility, Market Dynamics and Pricing Strategies Through ML

Developed the dynamic pricing strategies within the US aviation market using a 31GB Expedia dataset.

Estimated optimal ticket prices by leveraging distributed computing, advanced feature engineering, and non-linear machine learning models.

Gamma-Ray Bursts Calibrated using Artificial Neural Networks from the Pantheon+ Sample *Prof. Shantanu Desai*

Worked on the calibration of Amati relation of GRBs using ANNs and MCMC(Markov-Chain Monte Carlo) algorithm.

Movie Recommender System

Developed an interactive movie recommender system in Python using collaborative filtering techniques and large-scale user rating data, leveraging pandas and scikit-learn for data processing, model development, and personalized movie recommendations.

Credit Card Fraud Detection Model

Developed and deployed machine learning models in Python, including neural networks and ensemble classifiers, to accurately detect credit card fraud in highly imbalanced datasets, demonstrating data preprocessing, evaluation, and

visualization skills.

SKILLS

Languages C++/C,Python, SQL

Tools and Libraries Jupyter, Numpy ,Matplotlib, Pandas, Scikit-learn,Tensorflow, PyTorch, Excel, PySpark

RELEVANT COURSES

CSE: Data Structures and Algorithms, Introduction to Programming.

AI: Introduction to AI/ML, Probability and Random Variables.

Entrepreneurship: Business Analytics with python.

Mathematics: Linear Algebra, Statistics, Ordinary Differential Equations, Calculus.

Physics: Computational Physics, Data Science and Analysis, ML for High Energy Physics.

POSITIONS OF RESPONSIBILITY

Proactive UG Mentor | Sunshine - The counselling cell at IIT Hyderabad

UG Buddy | Sunshine

Mentored and guided 21 first-year students, earning the Star Performer Award for outstanding support and proactive engagement.

Hospitality and Logistics Coordinator | TEDxIITHyderabad

Managed the accommodation and logistics of the speakers of the flagship TEDxIITH event

Events and Outreach Core | Cepheid - The Astronomy/Astrophysics Club of IITH

Worked as a team and conducted workshops based on astronomy for 50+ school students.